

Causal contour deformation

FastSecDec · fixed and smooth dynamic strength

Working report · 10 October 2026 · measurements not yet complete

1 Deformation styles and formulae

1.1 Common geometry and fixed strength

After sector mapping, separate the real endpoint monomials from the smooth density. Let $F(x)$ denote the designated residual causal polynomial and let $U(x)$ denote each retained positive residual factor. The definition of F includes its physical scale: rescaling F also rescales this deformation. For a sampled sector with real masses, real kinematics and $x \in [0, 1]^n$ ($n \geq 1$), define

$$w_i = x_i(1 - x_i), \quad v_i = w_i \partial_i F, \quad z_{i(x)} = x_i - i\lambda(x)v_{i(x)}.$$

Exact zero-dimensional contributions bypass sampling. The coordinate faces remain fixed in their normal direction. With $A = \sum_i w_i (\partial_i F)^2 \geq 0$, the ray obeys

$$\text{Im } F(x - itv) = -tA + O(t^3).$$

Thus a sufficiently small positive strength moves the causal denominator to the lower half-plane. The scalar λ is a strength, not the norm of the displacement; no undocumented normalization by F or its gradient is used. Since cube coordinates are dimensionless, $[\lambda] = [F]^{-1}$; the same units apply to L . The displacement limit R is in parameter-space units.

Fixed mode takes a finite runtime value $\lambda(x) = \lambda_0 > 0$. Its Jacobian is

$$J_{ij} = \delta_{ij} - i\lambda_0 \partial_j v_i, \quad \partial_j v_i = \delta_{ij}(1 - 2x_i) \partial_i F + w_i \partial_j \partial_i F.$$

The complex, oriented determinant $\det J$ multiplies the density. A zero determinant alone is not a pole crossing. Neither its absolute value nor a frozen-gradient approximation gives the same contour integral.

For endpoint exponents $a_i + b_i \varepsilon$, the real endpoint monomials are retained and the smooth density receives

$$\det J \prod_i \rho_i^{a_i + b_i \varepsilon}, \quad \rho_i = \frac{z_i}{x_i} = 1 - i\lambda(x)(1 - x_i) \partial_i F.$$

The last expression defines ρ_i even at $x_i = 0$. Deform the numerator and residual factors as well. Apply this pullback **before** Taylor/IBP subtraction and Laurent construction. Boundary terms restrict the same full-sector map; they do not construct a new, lower-dimensional deformation.

1.2 Branches and checking policy

Continue each factor separately. Endpoint ratios have real part one; positive residuals must keep the appropriate branch, and the causal logarithm has the lower-lip value $\log(F - i0) = \log(|F|) - i\pi$ when $F < 0$. Combining these continued logarithms into the principal logarithm of their product can change the sheet. A stationary point with $F \neq 0$ is allowed. A surviving singularity at $F = 0$ and $A = 0$ requires an unresolved-deformation diagnostic, not a fabricated zero; a natively cancelled denominator or finite smooth density remains finite.

Optional validation has three policies: **always** checks pilot and production, **pilot** checks separate pilot work and removes optional production checks, and **off** removes both. Solver termination and nonfinite handling remain active. Fixed-mode failures identify the factor and point; they never silently shrink the requested strength. Finite pilots are safeguards, not global certificates. The gradient-map convention and subtraction ordering follow the established sector-decomposition approach; see pySecDec.

1.3 Dynamic strength: a smooth sufficient causal radius

Set $\lambda(x) = Sr(x)$ with $0 < S < 1$. The radius bounds the **whole homotopy** $x - itv(x)$ for $0 < t \leq Sr(x)$. It is not the distance to the nearest complex root: a positive real ray root need not exist, and a root-modulus prescription can vanish at the very physical pole one must avoid.

In the auxiliary ray expansion, hold $v(x)$ fixed. For odd $k \geq 3$, define

$$T_k = D^k F(x)[v^{k-2}, \cdot, \cdot], \quad B_k = \frac{\sum_{i,j} w_i w_j (T_k)_{ij}^2}{(k!)^2}.$$

For every positive residual U , write the real even-ray expansion

$$\operatorname{Re} U(x - itv) = U(x) + \sum_{k \geq 2, \text{ even}} d_{k(x)} t^k, \quad C_k = \frac{d_k^2}{U(x)^2}.$$

Native polynomial positivity admission supplies a strictly positive residual U on the closed cube; an arbitrary positive floor is not substituted. Let m_F and m_U count the structurally retained odd and even orders. With positive cap L and displacement scale R , solve

$$H(x, r) = \frac{r^2}{L^2} + r^2 \frac{\|v\|^2}{R^2} + m_F \sum_{k \geq 3, \text{ odd}} B_k r^{2k-2} + \sum_U m_U \sum_{k \geq 2, \text{ even}} C_k r^{2k} = 1.$$

All coefficients are nonnegative. The cap term makes the positive root unique and simple, with $r \leq L$ and $rH_r \geq 2$ at the root. Both constraints are smooth; there is no hard minimum whose derivative would jump.

Why the bound is causal. Put $W = \operatorname{diag}(w_i)$ and $q = \sqrt{W} \operatorname{grad} F$, so $A = q^T q$ and $v = \sqrt{W} q$. The higher odd ray term satisfies

$$\left| D^k \frac{F[v^k]}{k!} \right| \leq A \sqrt{B_k}, \quad \sum_k \sqrt{B_k} t^{k-1} \leq \sqrt{m_F \sum_k B_k t^{2k-2}}.$$

Consequently higher imaginary terms cannot cancel the leading $-tA$ along the admitted homotopy when $A > 0$. The even-factor bound similarly keeps $\operatorname{Re} U(z) > 0$. The proof does not divide by A , so small gradients do not create an artificial singular bound. At $A = 0$, the stationary-point rules still apply.

1.4 Full local-strength derivatives

Dynamic strength changes the Jacobian:

$$J_{ij} = \delta_{ij} - i(\lambda \partial_j v_i + v_i \partial_j \lambda).$$

Implicit differentiation of $H(x, r(x)) = 1$ gives

$$\partial_j r = -\frac{\partial_j H}{\partial_r H}, \quad \partial_j \lambda = S \partial_j r.$$

Here $\partial_j H$ holds r fixed but differentiates all physical coefficient dependence, including v and the smooth constraints. Symbolic endpoint reduction differentiates the complete deformed density, including this identity and the determinant. Solver iteration choices are never differentiated. Every face retains the full-sector dimension, structural counts and cap parameters. Omitting the rank-one term $-iv(\operatorname{grad} \lambda)^T$ changes the integral.

The polynomial construction is an independently checked baseline. Its exact-arithmetic proof is distinct from numerical validation of a rounded implementation. Disabling optional checks removes their runtime cost, not solver failures or nonfinite-value diagnostics.

1.5 Sign-aware refinement of the dynamic bound

The baseline bounds beneficial as well as harmful higher terms. A smoother, tighter alternative bounds only terms that can oppose causality. Define the real symmetric matrix

$$R_k = \frac{(-1)^{\frac{k+1}{2}} L^{k-1}}{k!} \sqrt{W} T_k \sqrt{W}.$$

It need not be built with explicit square roots of W . Weighted trace and squared-norm identities provide the native polynomial quantities

$$a_k = \frac{\text{tr}(R_k)}{n}, \quad b_k^2 = \frac{n-1}{n} \left(\text{tr}(R_k^2) - \frac{\text{tr}(R_k)^2}{n} \right).$$

For a symmetric matrix its largest eigenvalue is at most $a_k + \sqrt{b_k^2}$. Smooth both the spectral bound and its positive part with a fixed dimensionless $\delta = 10^{-3}$:

$$t_k = a_k + \sqrt{b_k^2 + \delta^2}, \quad p_{\delta(t)} = \frac{t + \sqrt{t^2 + \delta^2}}{2}, \quad \mu_k = p_{\delta(t_k)}.$$

For negative t , evaluate the exactly equivalent expression

$$p_{\delta(t)} = \frac{\delta^2}{2(\sqrt{t^2 + \delta^2} - t)}$$

to avoid cancellation. For each even U coefficient, use

$$\nu_{U,k} = p_{\delta(-d_k \frac{L^k}{U})}.$$

With $u = \frac{r}{L}$, form

$$E_{F(u)} = \sum_{k \geq 3, \text{ odd}} \mu_k u^{k-1}, \quad E_{U(u)} = \sum_{k \geq 2, \text{ even}} \nu_{U,k} u^k.$$

Then solve the unique positive root

$$u^2 + \frac{L^2 u^2 \|v\|^2}{R^2} + E_{F(u)}^2 + \sum_U E_{U(u)}^2 = 1, \quad \lambda = S L u.$$

The nonnegative envelope coefficients retain monotonicity and smoothness. Their derivatives, including smoothing, enter the same implicit-jet machinery. A larger permitted displacement is a diagnostic; it does not establish lower variance or lower time to precision. The baseline and sign-aware construction must be compared with their full evaluation costs.

1.6 Sampling, diagnostics and interpretation

Korobov remains an outer real map: $y = T(q)$ followed by $z = z(y)$. Its total Jacobian is $\det J_{z(y)} \prod_i T_{i'}(q_i)$, with each factor applied once. No inverse Korobov derivative is introduced at an endpoint. Contour evaluation is deterministic and consumes no production random streams.

QMC uncertainties come from complete independently randomized lattices; democratic QMC retains covariance between sectors sharing a shift. Discrete Havana MC retains its frozen production proposal and uses complete batches. Both accumulate the full Laurent vector and its covariance. A partial allocation, a missing sector or an unvisited zero is not exact evidence.

The smooth sufficient-radius bounds above are FastSecDec's parameter-space construction. They are not attributed to pySecDec or to the LTD algorithm. Related work includes Winterhalder et al., *Targeting Multi-Loop Integrals with Neural Networks* on optimized contour choices, and Capatti et al., *Numerical Loop-Tree Duality: contour deformation and subtraction* on causal contour construction and physical multiloop tests. Their timings are not directly comparable FastSecDec speed measurements.

2 Physical $gg \rightarrow hh$: the 1000 GeV double box

The input is the existing individual **D05** s-channel double-box contribution, with incoming ++ helicities, $\sqrt{s} = 1000$ GeV, $\cos(\theta) = \frac{4}{5}$, $m_H = 125$ GeV and $m_t = y_m = 172.5$ GeV. HEPKit supplies the native graph, contracted numerator, wavefunctions and kinematics. The colour contraction is unnormalized δ_{ab} ; the measure is $\prod_l d^D \frac{k_l}{i\pi^{\frac{D}{2}}}$, $D = 4 - 2\varepsilon$, with multiplier one. This is not a complete diagram sum or a spin/colour averaged matrix element.

Both prescriptions use symbolic endpoint IBP. Fixed mode uses the symbolic Jacobian; dynamic mode uses contour-only dual evaluation. Both complete artifacts contain 30 sampled sectors and 30 exact records.

2.1 Complete generation

Contour	Workers	Minutes	Peak GB	Saved MB	Exact / JIT MB
Fixed	6	43.57	54.09	70.41	3.20 / 4.66
Dynamic	$2 \rightarrow 4$	106.81	60.94	51.34	11.33 / 15.98

Both use serial generation, SymJIT O2, Horner zero and common-pair cap 1000. Dynamic time includes the initial two-worker attempt and its four-worker continuation; the 26.1-second restart gap is excluded. Initial native series widths are one and two, respectively. These are complete-campaign observations, with different worker counts; the matched Jacobian comparison is in section 3. GB and MB are decimal; RSS includes the owned parent and child processes.

2.2 Five minutes on 50 physical cores

Method	Contour	Finite coefficient	Joint variance	Relative SE
QMC	Fixed	$68.35 + 26.24i$	2.46049	2.142%
QMC	Dynamic	$69.45 + 24.11i$	7.8327	3.807%
Discrete MC	Fixed	$69.07 + 24.93i$	0.256372	0.690%
Discrete MC	Dynamic	$68.45 + 24.65i$	0.500102	0.972%

For the finite complex estimate $\hat{I}_0$, the joint estimator variance is $V = \text{tr } \Sigma_0$ and relative joint standard error is $\frac{\sqrt{V}}{|\hat{I}_0|}$. This is not pointwise integrand variance. The reproducibility JSON retains both diagonal covariance entries and the off-diagonal entry, all Laurent components, exact offsets, work counts and precision-rescue diagnostics.

Method	Contour	Seconds	Peak GB	Sample μs avg / max
QMC	Fixed	302.32	1.94	3.73 / 4.02
QMC	Dynamic	302.37	2.02	16.07 / 28.97
Discrete MC	Fixed	303.53	4.13	23.99 / 43.54
Discrete MC	Dynamic	302.27	11.31	71.79 / 164.48

Sample costs are the average and maximum of per-sector mean integrand costs, including precision rescue and amortized setup/adaptation work; they are not maximum single-point latency. A separate fixed all-sector probe measured 10.44 μs average and 14.06 μs maximum sector mean on a smaller fixed point set.

Fixed uses $\lambda = 10^{-6}$; dynamic uses $S = .8$, $L = 10^{-6}$, $R = 1$, selected with separate tuning seeds. Dynamic/fixed variance ratios are 3.18 (QMC) and 1.95 (MC): fixed performs better in these runs. Each row is one independent production run; these ratios have no repeated-run confidence interval. The 300-second native integration budget includes worker setup and Havana adaptation. Loading, binding and causal Pilot16 precede that clock; optional production causal checks are disabled. Complete accepted work is drained after cancellation, explaining the modest overshoot. All rows retain full covariance and independently reserved production streams; incomplete work is not added. No independent numerical reference exists for this point. These frozen campaign executables precede the loader-cache update in section 3.

3 Implementation and performance

3.1 Preserve algebraic structure and reuse native facilities

FastSecDec owns the sector/integration interfaces in Rust. Symbolica supplies factored atoms, differentiation, function maps, evaluator optimization, duals, and serialization; Numerica supplies arithmetic, matrices, RNG primitives and Havana. HEPKit and Linnet retain graph, model and kinematic ownership. HEPKit's Python bridge wraps native objects rather than duplicating algorithms.

The smooth density is deformed before subtraction. Native function definitions can share repeated Jacobian and envelope bodies while preserving their derivatives and explicit radius dependencies. The saved program restores the optimized evaluator; binding a strength or cap must not rerun Horner/CPE. SymJIT O2 remains the native backend default; portable exact evaluator IR is retained separately for eager and high-precision execution.

An actual D05 source sector contains 236 smooth terms. The initial representation copied its large Jacobian into each term. Native structured construction plus a shared determinant body reduced that sector's mapped record from 683.8 MB to 365.6 MB and sampled peak RSS from 2.178 GB to 1.288 GB. Mapping wall time was 10.93 s versus 7.95 s on a shared host. In the native debug graph, compilation fell from 169.49 s to 58.16 s. Saved full-vector parity passes at two admitted points; both representations reject the same excessive strength. Saved program size increases by 6.6%. These are one-sector measurements, not full-generation timings or a convergence result.

3.2 Symbolic versus dual determinant

The symbolic route constructs a native determinant and optimizes it together with the density. It offers common-expression elimination across outputs but may make generation expensive when a large body is repeatedly substituted. The first optional dual route also applied numerical-dual subtraction to the whole density. It coupled the two choices, contrary to the clarified requirement. Its historical measurements include high-order endpoint jets and cannot establish the cost or limits of the corrected contour-only construction.

Independent contour-only construction. Endpoint reduction and subtraction remain symbolic in both variants. Native differentiation retains every bulk, boundary, determinant and local-strength derivative. At final compilation, native exact-subtree replacement supplies surviving first-image derivatives from a degree-one image-only Dualizer; higher endpoint derivatives are already symbolic. Coordinate seeds remain one on subtraction faces, and parameter seeds remain zero. Native pruning removes unused inputs before composition. Exact-only boundary contributions retain symbolic evaluation and cancellation.

Five focused compiler controls pass, including third-order endpoint poles with fixed and both dynamic prescriptions, saved programs, unused callbacks and private-input collisions. Public Symbolic IBP controls also pass with both implicit-radius prescriptions, complete complex Laurent vectors and restored programs. Conditional bodies requiring hoisted image inputs are explicitly unsupported by this opt-in route. The matched source-sector comparison below includes generation, RSS, program sizes and sampling cost. Full-integral generation and all four physical integration runs have completed. No mathematical reordering, alternative CAS or AD implementation is introduced. An optional initial native Series width avoids a known discarded attempt; the strict absolute Laurent-remainder coverage check and checked retries remain unchanged. It defaults to one. Twenty-three focused controls and independent native-reuse reviews pass; production savings are still being measured.

3.3 Matched Jacobian measurements on one source sector

Contour / Jacobian	Time / peak RSS	Exact / JIT IR	Mean sample
Fixed / symbolic	559.9 s / 12.01 GB	108.3 / 156.2 kB	14.64 μ s
Fixed / dual	566.9 s / 12.01 GB	136.8 / 202.8 kB	19.20 μ s
Dynamic / symbolic	1511.6 s / 28.44 GB	294.0 / 524.6 kB	37.71 μ s
Dynamic / dual	794.9 s / 16.73 GB	372.0 / 520.0 kB	29.99 μ s

Times include generation, compilation, saving and pointwise checks in separate processes; sizes describe evaluator IR, not the complete saved owner. Settings are SymJIT O2, Horner zero and common-pair cap 1000. Each matched pair uses 8,192 identical points: fixed $\lambda = 10^{-6}$ and dynamic $S = .8$, $L = 10^{-6}$, $R = 1$. Both fixed arms rescue 27 points in double-double; both dynamic arms rescue 28. Fresh-process restoration and complete-vector parity pass. This sector favors symbolic J for fixed mode and dual J for dynamic mode.

3.4 Residency, callback ownership and statistical work

Universal indexed artifacts let serial generation publish one completed sector and release its worker process. Selective loading retains optimized native programs. Ordinary discrete MC uses caller-owned parallel execution; the 50-core

campaign must account for all resident evaluator contexts. Aggregate parent-and-child RSS measures actual residency, including JIT scratch.

Independent probes exposed a native JIT-clone callback issue: cloned evaluator tables were not used by machine code, which retained the original callback environment and serialized its mutable workspace. The narrow owner correction, Symbolica PR 62, is retained in public consumer revision 74225696. The same consumer adds Symbolica PR 63: common-subexpression lookup remaps operands before hashing them, avoiding one optimizer pass per level of a duplicate dependency chain. Native branch ancestry and callback argument order remain unchanged. The current workspace passes 935 tests (34 explicit ignores, including a child invoked separately). The portable suite passes 83 tests. Strict Clippy, formatting, and thin-binding build/lint checks pass. These are native/portable-host gates, not installed Python-wheel or browser/WASM execution evidence.

The physical QMC run also exposed avoidable coordinator work: matching common shifts used linear searches within already-sorted native replica records. Binary search preserves the selected rows and arithmetic order. Cancellation now drains dispatched results without repeating expensive observational reductions and checkpoint writes for every return; the final native checkpoint and report still include accepted returns. Deterministic cancellation and reordered, gapped-shift covariance/restoration controls cover these changes.

3.5 Persist the native primary evaluator

The old artifact saved exact evaluator IR but discarded the compiled primary. The new record also saves Symbolica's native JIT payload. Compatible loads restore its lowered application and callback descriptors; the owner still emits machine code. Exact IR remains available for precision rescue and portable execution. Existing mathematical identities and caller-owned scheduling remain unchanged. Immutable exact templates and artifact buffers are shared across workers; mutable stacks and callback environments remain independent.

On two preserved source-sector records, matched optimized cache-miss/cache-hit loads measured 0.1804/0.1437 s (fixed) and 0.3865/0.3025 s (polynomial). The optional cache adds 13.34% to those records. These are loader measurements on historical endpoint constructions, not new physical integration results. Changing dynamic validation factories also reuses the native primary instead of rebuilding it from exact IR. Optional byte-integrity checks remain optional; format, compatibility and native decoding checks are retained.

A second probe restores the same complete 70.41 MB fixed artifact in the old and new optimized implementations, then constructs 50 whole-owner worker clones. Post-clone RSS falls from 6.68 to 1.79 GB, and clone time from 3.32 to 0.78 s. Loading the original uncached artifact remains similar, 3.81 versus 3.71 s. This measures immutable ownership improvements without binding, pilot or sampling work; it is separate from the cache-hit comparison above.

Stream identities are coordinator-owned. Pilots, production replicas and retries keep separate identities; rejected, duplicate or stale work cannot enter the estimate. Optional causal checks are truly absent in unchecked production, while solver/nonfinite failures remain visible. Accepted common QMC shifts and complete MC batches update the full covariance, preserving sector cancellations even at a time-budget stop.

Reproduction and source: FastSecDec, `contour_deformation`. Native input: `examples/contour/gghh_double_box_1000/`. Mathematical audit: `docs/reviews/contour-foundation.md`. Physical campaign provenance and complete covariance are in `docs/contour-deformation-results.json`; the scalar extension is in progress.